

Sensitivity analyses for positivity violations

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¹Footnotes are for asides or references

Motivating Example

Question: What is the mean systolic blood pressure (SBP) in mm Hg among children and adolescents aged 2-17 in the United States?

Parameter: $\mu = E[Y]$ where Y_i is SBP, and $E[\cdot]$ is the expected value function²

Data: NHANES 2017-2018

Systematic Error: SBP is missing for 44% of children

²Can think about $E[\cdot]$ as $1/n \sum_{i=1}^n (\cdot)$

Extent of missingness means that estimate is sensitive to choices about missing data

For example, a complete-case analysis assumes

$$E[Y] = E[Y \mid R = 1]$$

where R_i denotes an indicator of SBP being measured

A Weaker Assumption

Instead of a complete-case analysis, we can assume that SBP is missing completely at random conditional on X (i.e., conditionally exchangeable)

$$E[Y \mid X = x] = E[Y \mid X = x, R = 1] \text{ for all values of } x$$

Can freely condition on R

Under this assumption, μ is identified by³

$$\sum_{x \in \mathcal{X}} E(Y \mid X = x, R = 1) \Pr(X = x)$$

or a weighted average of the complete-case means within strata of X

³Estimate with direct maximum likelihood estimator, g-computation

Exchangeability necessitates a *positivity* assumption

Definition:

The diagram shows the equation $\Pr(R = 1 \mid X = x) > 0 \quad \forall x \in \mathcal{X}$ with three colored background regions: light blue for the probability term, light orange for the greater-than-zero symbol, and light purple for the universal quantifier and its domain. Annotations include: a blue arrow from 'Probability of Y being observed' to the probability term; an orange arrow from 'is non-zero' to the greater-than-zero symbol; and a purple arrow from 'for all unique values of X' to the universal quantifier and its domain.

$$\Pr(R = 1 \mid X = x) > 0 \quad \forall x \in \mathcal{X}$$

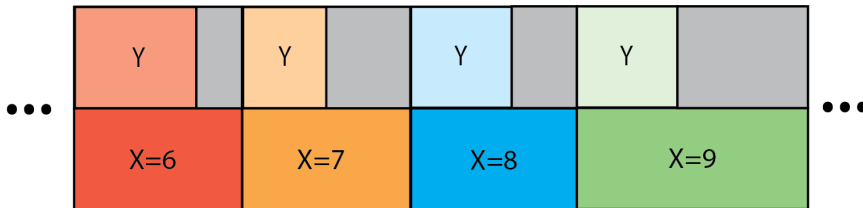
Annotations:

- Probability of Y being observed (points to $\Pr(R = 1 \mid X = x)$)
- is non-zero (points to > 0)
- for all unique values of X (points to $\forall x \in \mathcal{X}$)

Intuitively, we need some individuals with $X = x$ to have Y observed at least sometimes so can 'stand-in' for those without Y measured

Positivity for SBP in NHANES

If X is age, then positivity can be depicted as

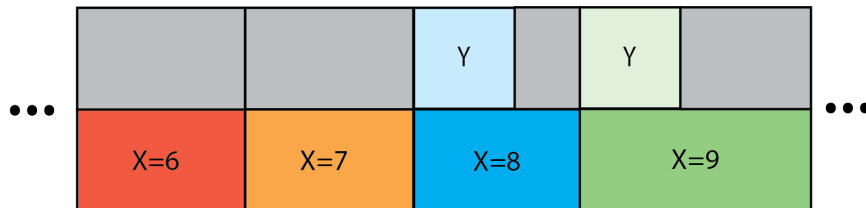


Every unique age in the population has some individuals with SBP measured

Nonpositivity by Age in NHANES

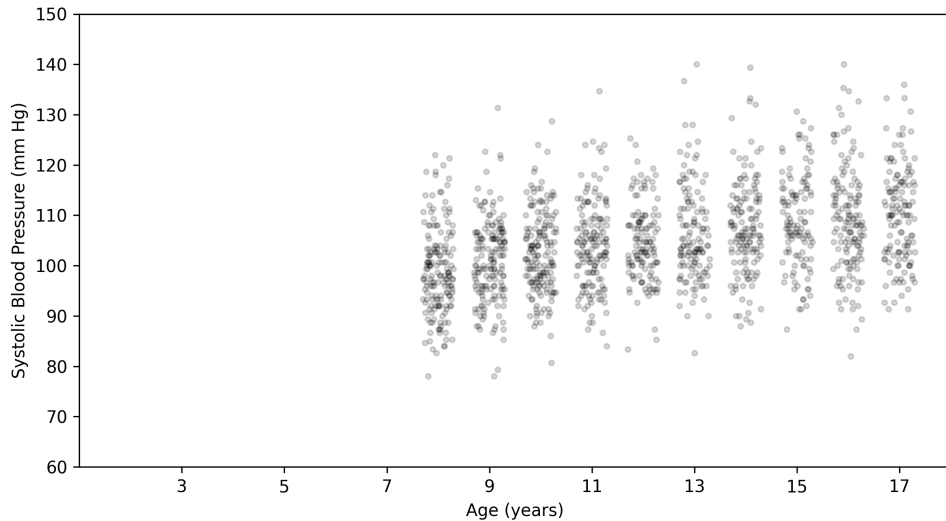
In NHANES 2017-2018,

- “BP is measured on participants 8 years and older”⁴



⁴https://wwwn.cdc.gov/Nchs/Nhanes/2017-2018/BPX_J.htm

NHANES Data



Addressing Positivity Violations

Consider two routine methods for dealing with positivity violations

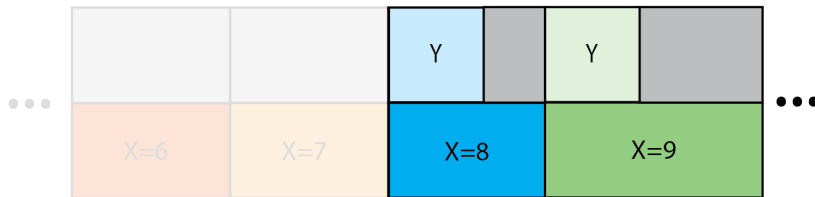
1. Change the parameter
2. Extrapolate from a model

Option 1: Change the Parameter

Parameter (revised): mean SBP for those 8 or older

$$\mu^* = E[Y \mid X \geq 8]$$

Question: what is the mean systolic blood pressure (SBP) in mm Hg among children and adolescents aged 8-17 in the US?

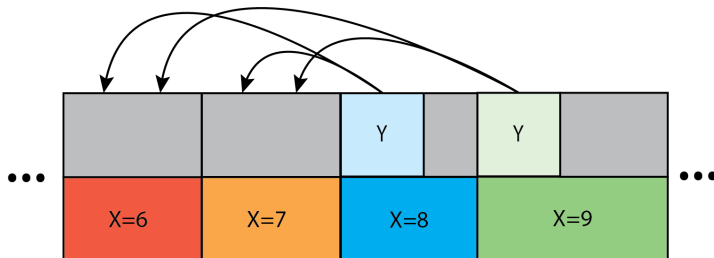


Option 2. Extrapolate

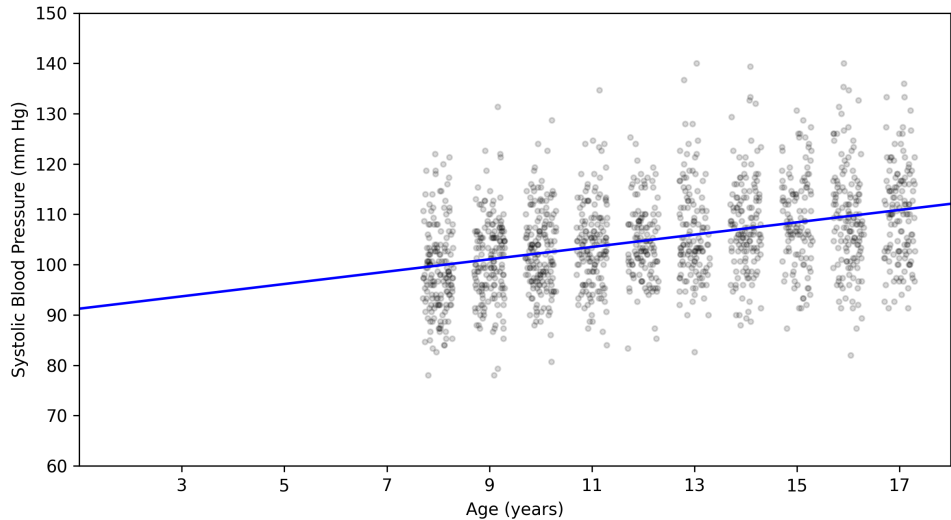
Fit a parametric model

$$E[Y \mid X \geq 8, R = 1] = \beta_0 + \beta_1 X$$

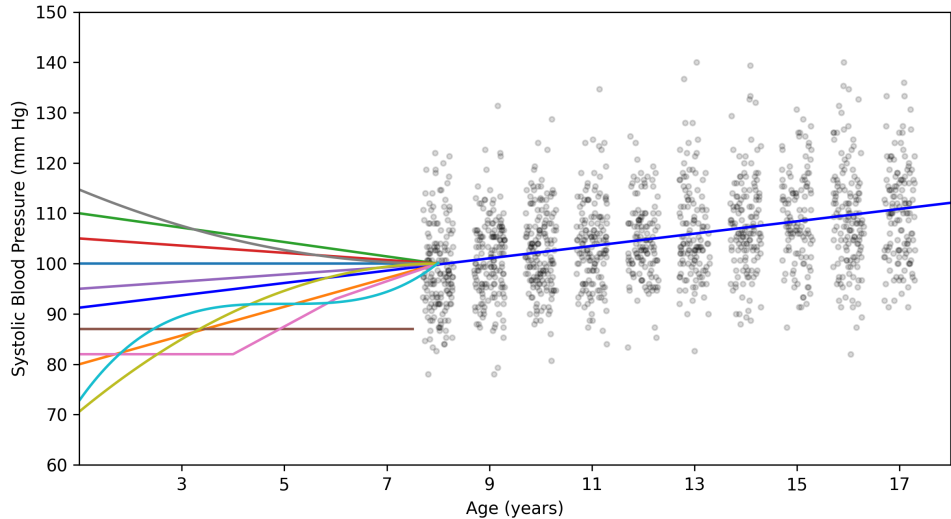
then fill in missing SBP with model predictions



Extrapolation for NHANES



The Issue with Extrapolation



An Alternative Approach

Divide parameter of interest in parts

$$\mu = \overset{\text{Overall mean}}{\mu} = \overset{\text{Mean in nonpositive region}}{E[Y \mid X < 8] \Pr(X < 8)} + \overset{\text{Mean in positive region}}{E[Y \mid X \geq 8] \Pr(X \geq 8)}$$

Strategy: divide and conquer

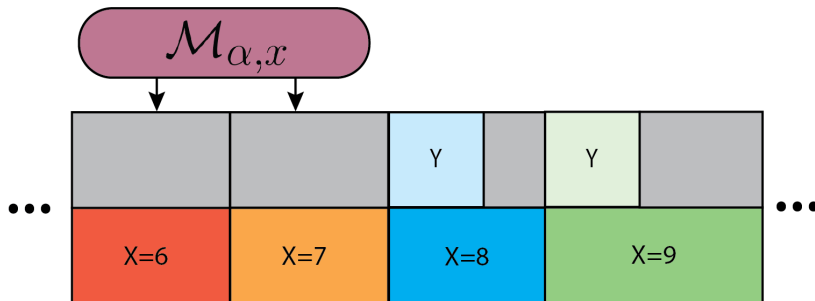
Here, we can still identify the mean in the positive region

$$\mu^* = E[Y \mid X \geq 8] = \sum_{x \in \mathcal{X}} E(Y \mid R = 1, X = x, X \geq 8) \Pr(X = x \mid X \geq 8)$$

Dealing with Non-Positivity

Available data leaves us stuck with $E[Y | X < 8] = ?$

- To progress, will appeal to external information



At the most basic level, we know that SBP must be a (finite) positive number

$$E[Y \mid X = x] \in (0, c) \text{ where } c < \infty \text{ for ages 2-7}$$

Context allows us to further narrow the plausible region to

$$E[Y \mid X = x] \in [50, 140] \text{ for ages 2-7}$$

as these boundary values would constitute medical emergencies

Following some algebra, we can show that μ must be greater than

$$50 + \Pr(X \geq 8) \left\{ E[Y \mid X \geq 8] - 50 \right\}$$

and less than

$$140 + \Pr(X \geq 8) \left\{ E[Y \mid X \geq 8] - 140 \right\}$$

Sensitivity Analysis

In this example, the bounds are quite wide, $[85.8, 116.8]$

Rather than evaluate at the extremes, evaluate

$$\mu = \sum_{x \in \mathcal{X}} E(Y \mid R = 1, X = x, X \geq 8) \Pr(X = x \mid X \geq 8) \Pr(X \geq 8) + q \Pr(X < 8)$$

where q is set to a particular value

Some readers might be more optimistic than others

- Rather than select one q for everyone, can show how μ varies as a function of q

Our algebra before gave us the following formula

$$q + \Pr(X \geq 8) \left\{ E[Y | X \geq 8] - q \right\}$$

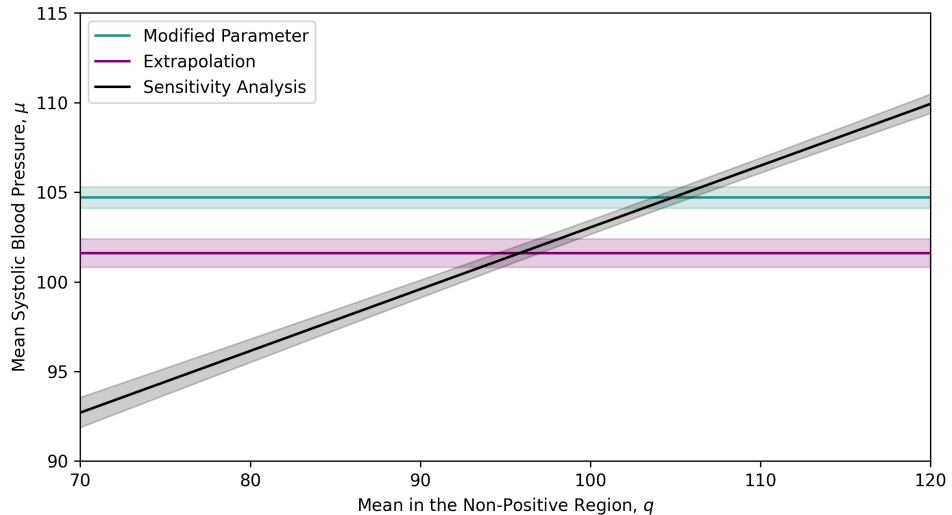
I will use g-computation to estimate $E[Y | X \geq 8]$,⁵ but could use IPW or AIPW

95% Confidence intervals are computed using the sandwich variance with q held fixed⁶

⁵Cole et al. *Am J Epidemiol*, 2023; 192(1):6-10

⁶Ross et al. *Int J Epidemiol* 2024; 53(2):dyae030

Results by Method



Conclusions

Positivity is often neglected as an assumption

- But it remains vital for identification of many parameters

Restricting parameters to positive regions may be reasonable

- But may limit utility of research
- Extrapolation premised on strong assumptions

Approach described here (1) retain interest parameter, and (2) less strict assumptions

Replace fixed q with a distribution

- Can further replace q with a mathematical model to incorporate more rich external information⁷

Exchangeability (and thus positivity) is used to address other systematic errors

- Approach can be used to address nonpositivity in these settings⁸

Approach based on knowledge of the X where non-positivity occurs⁹

⁷Some work on formalizing identification with mathematical models: Zivich *arXiv:2511.01960*

⁸Originally proposed in the context of transporting trials: Zivich et al. *Epidemiology* 2024;35(1):23-31, Zivich et al. *JRSSA* 2025;188(1):158-180

⁹Susmann et al. *arXiv:2509.20206* have an approach based on the propensity score that does not require this knowledge to construct corresponding bounds

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Appendix

Following definition of conditional expectation

$$E[Y \mid X = x, R = 1] = \frac{E[Y R I(X = x)]}{\Pr(R = 1 \mid X = x) \Pr(X = x)}$$

So $\Pr(R = 1 \mid X = x)$ results in a division by zero when positivity is not met

Synthesis of Statistical and Mathematical Models

